

Diagnostic report: Mg I

Computed levels: 25

Matched to NIST: 15

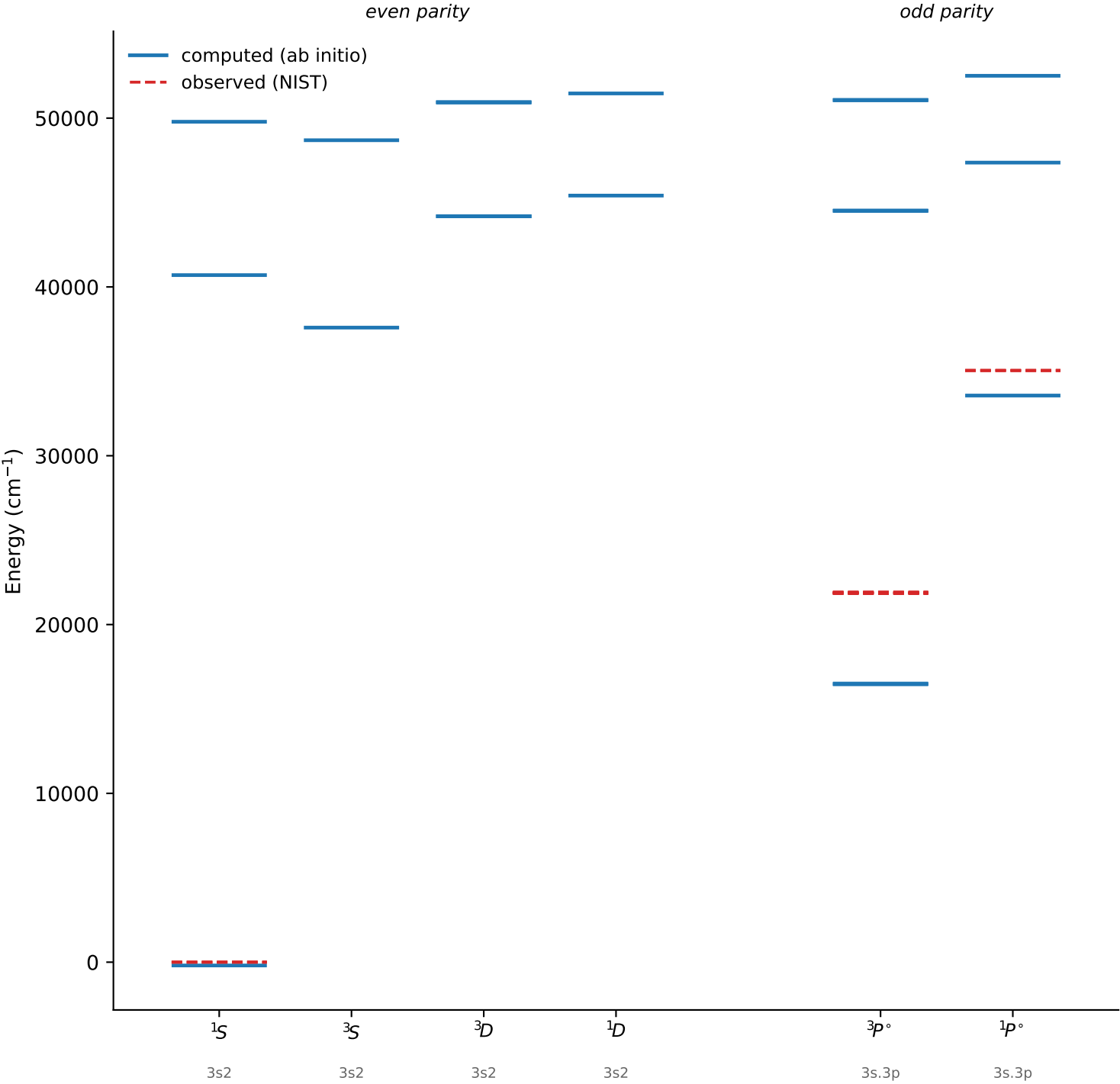
Computed E1 lines: 52

Pre-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 24186 cm^{-1} (ab initio, no RCE fit yet)

Strongest computed lines:

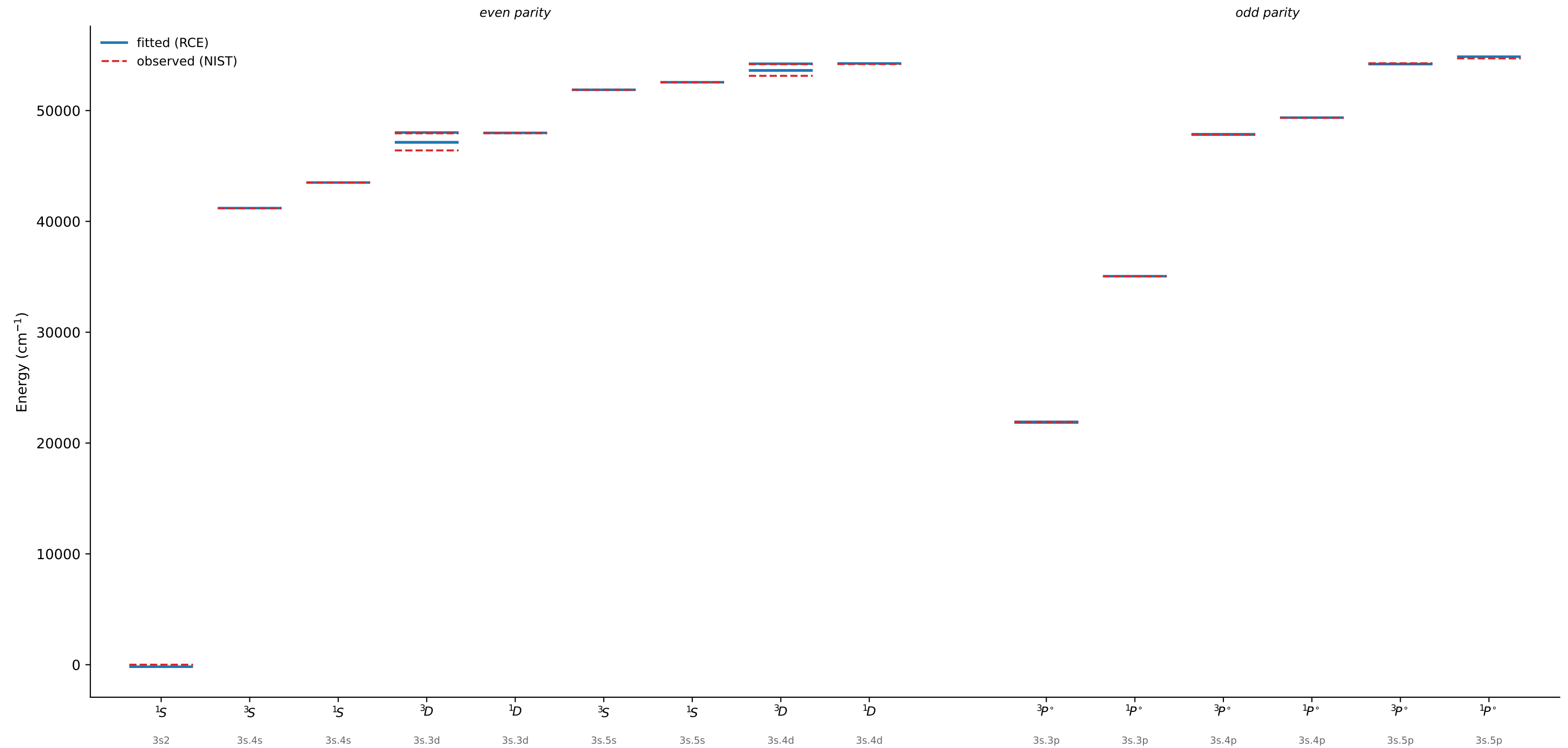
lambda(A)	log gf	upper term
24404.97	+0.595	(2S) 1P
42031.78	+0.507	(2S) 3P
3612.66	+0.470	(2S) 3P
15581.33	+0.424	(2S) 3P
8441.88	+0.421	(2S) 1P
2962.04	+0.393	(2S) 1P
14430.30	+0.382	(2S) 3P
42069.26	+0.285	(2S) 3P

Mg I: term diagram - ab initio vs observed



are terms (grouped by parity), ticks are J-levels. Solid = computed (ab initio), dashed = observed (NIST). Per-level energies and residuals are tabulated

Mg I: term diagram - fitted vs observed



Columns are terms (grouped by parity), ticks are J-levels. Solid = fitted (RCE), dashed = observed (NIST). Per-level energies and residuals are tabulated later.

Mg I: levels — ab initio and fitted vs observed (cm⁻¹)

config	term	J	E_obs	E_calc	Δ_abinit	E_fit	Δ_fit
3s2	¹ S	0	0.0	-195.0	-195.0	-183.1	-183.1
3s2	¹ S	0	0.0	40700.0	+40700.0	-183.1	-183.1
3s2	¹ S	0	0.0	49785.0	+49785.0	-183.1	-183.1
3s.3p	³ P°	0	21850.4	16463.0	-5387.4	21850.3	-0.1
3s.3p	³ P°	1	21870.5	16477.0	-5393.5	21870.5	+0.0
3s.3p	³ P°	2	21911.2	16505.0	-5406.2	21911.1	-0.1
3s.3p	³ P°	0	21850.4	44511.0	+22660.6	21850.3	-0.1
3s.3p	³ P°	1	21870.5	44514.0	+22643.5	21870.5	+0.0
3s.3p	³ P°	2	21911.2	44519.0	+22607.8	21911.1	-0.1
3s.3p	³ P°	0	21850.4	51066.0	+29215.6	21850.3	-0.1
3s.3p	³ P°	1	21870.5	51067.0	+29196.5	21870.5	+0.0
3s.3p	³ P°	2	21911.2	51070.0	+29158.8	21911.1	-0.1
3s.3p	¹ P°	1	35051.3	33566.0	-1485.3	35053.2	+1.9
3s.3p	¹ P°	1	35051.3	47365.0	+12313.7	35053.2	+1.9
3s.3p	¹ P°	1	35051.3	52508.0	+17456.7	35053.2	+1.9
3s2	³ S	1	--	37590.0	--	--	--
3s2	³ S	1	--	48690.0	--	--	--
3s2	³ D	1	--	44185.0	--	--	--
3s2	³ D	2	--	44185.0	--	--	--
3s2	³ D	3	--	44185.0	--	--	--
3s2	³ D	1	--	50937.0	--	--	--
3s2	³ D	2	--	50937.0	--	--	--
3s2	³ D	3	--	50937.0	--	--	--
3s2	¹ D	2	--	45411.0	--	--	--
3s2	¹ D	2	--	51462.0	--	--	--

Mg I: level residuals (ab initio RMS = 24186, fit RMS = 272 cm⁻¹)

